

Jardim's exercise sessions

UFBA, Salvador, September 28–October 1, 2026

1 Sections, Rao modules and linkage

Can two curves be joined by a chain of links? The exercises ask for more: can they come from sections of twists of one common rank-two bundle?

Work on $X = \mathbb{P}_{\mathbb{C}}^3$. Write $\mathcal{O}(a) = \mathcal{O}_X(a)$, $F(a) = F \otimes \mathcal{O}(a)$ and $H_*^i(F) = \bigoplus_t H^i(X, F(t))$.

For a rank-two bundle E with $\det E = \mathcal{O}(c)$, a section $s \in H^0(E(a))$ whose zero scheme is a curve C of pure codimension two gives

$$0 \longrightarrow \mathcal{O}(-a) \xrightarrow{s} E \longrightarrow \mathcal{I}_C(c+a) \longrightarrow 0. \quad (1)$$

Locally $s = (f, g)$ and $C = V(f, g)$. Two equations of codimension two in the smooth space X form a regular sequence: thus C is locally a complete intersection. Consequently the map $(u, v) \mapsto fv - gu$ has image (f, g) and kernel generated by s . Determinants give the twist $c + a$. The inclusion is *saturated*: its quotient is torsion-free.

Which curves arise this way? Exactly the locally complete intersection curves with $\omega_C \cong \mathcal{O}_C(m)$ for some integer m (*subcanonical* curves), by Hartshorne–Serre [1]. Even the smooth twisted cubic fails: $\deg \omega_C = -2$, whereas $\deg \mathcal{O}_C(m) = 3m$.

The *Rao module* of a locally Cohen–Macaulay curve is

$$M_C = H_*^1(\mathcal{I}_C).$$

It records sections on C that do not extend to ambient polynomials, together with multiplication by polynomials. The curve is arithmetically Cohen–Macaulay precisely when $M_C = 0$.

Rao's theorem. For *all* locally Cohen–Macaulay curves in X , isomorphism of Rao modules up to shift is equivalent to belonging to the same *even liaison class* [2]. A liaison is a chain of complete-intersection links; “even” counts the links $(2, 4, \dots)$, not the rank of a bundle. No presentation by a rank-two bundle is required by the theorem.

What a common bundle gives. Since $H^1(\mathcal{O}(m)) = H^2(\mathcal{O}(m)) = 0$ for every m , two sequences (1) with the same E give

$$M_{C_1}[c + a_1] \cong H_*^1(E) \cong M_{C_2}[c + a_2], \quad N[e]_t = N_{t+e}.$$

Thus a common bundle implies even liaison. The converse fails: a line and a smooth conic already give a counterexample (Section 2).

Why introduce sections? They can produce the geometry of links, not just an existence criterion. For sections independent at the generic point, put $A_i = \mathcal{O}(-a_i)$ and $T = E/(A_1 + A_2)$, identifying each A_i with its image. The determinant defines a surface $S = Z(s_1 \wedge s_2)$ containing both curves. Quotienting in either order gives

$$\mathcal{I}_{C_1/S}(c + a_1) \cong T \cong \mathcal{I}_{C_2/S}(c + a_2).$$

If S is smooth, this says $C_2 \sim C_1 + (a_2 - a_1)H_S$, where H_S is its hyperplane class. Choose a general $D \sim mH_S - C_1$ for m large. Then both $C_1 + D$ and $C_2 + D$ are hypersurface sections of S : this gives $C_1 \leftrightarrow D \leftrightarrow C_2$, two links. For $S = V(F)$ and $C_1 \cup D = V(F, G)$, the residual ideal is $I_D = (F, G) : I_{C_1}$.

2 A line and a smooth conic

Let L be a line and Q a smooth conic. Their ideal resolutions are

$$\begin{aligned} 0 \rightarrow \mathcal{O}(-2) \rightarrow \mathcal{O}(-1)^{\oplus 2} \rightarrow \mathcal{I}_L \rightarrow 0, \\ 0 \rightarrow \mathcal{O}(-3) \rightarrow \mathcal{O}(-1) \oplus \mathcal{O}(-2) \rightarrow \mathcal{I}_Q \rightarrow 0. \end{aligned}$$

Hence both Rao modules vanish: they are in the same even liaison class. Individually they are zero schemes of sections of $\mathcal{O}(1)^{\oplus 2}$ and $\mathcal{O}(1) \oplus \mathcal{O}(2)$.

A common rank-two bundle is impossible. Adjunction for a regular section gives $\omega_C \cong \mathcal{O}_C(c + 2a - 4)$. On L this forces $c + 2a_1 = 2$; on Q it forces $c + 2a_2 = 3$. Their difference would be $2(a_2 - a_1) = 1$. The same Rao module does not enforce the determinant needed by a common bundle.

A common reflexive sheaf forces coplanarity. Suppose instead that a rank-two reflexive E fits into

$$\begin{aligned} 0 \rightarrow \mathcal{O}(-a_1) \xrightarrow{s_1} E \rightarrow \mathcal{I}_L(c + a_1) \rightarrow 0, \\ 0 \rightarrow \mathcal{O}(-a_2) \xrightarrow{s_2} E \rightarrow \mathcal{I}_Q(c + a_2) \rightarrow 0. \end{aligned}$$

Put $h = c_1(\mathcal{O}(1))$. The resolutions above give

$$\begin{aligned} c(\mathcal{I}_L(t)) &= 1 + th + h^2 + (2 - t)h^3, \\ c(\mathcal{I}_Q(t)) &= 1 + th + 2h^2 + (6 - 2t)h^3. \end{aligned}$$

Multiply by $1 - a_i h$ and compare the h^2 coefficients for E :

$$1 - a_1(c + a_1) = 2 - a_2(c + a_2), \quad 1 = (a_2 - a_1)(c + a_1 + a_2).$$

Meanwhile

$$s_1 \wedge s_2 \in H^0(\mathcal{O}(d)), \quad d = c + a_1 + a_2.$$

The determinant is nonzero. The integer equation gives $a_1 \neq a_2$; choose i, j with $a_i < a_j$. If $s_1 \wedge s_2 = 0$, the composite $\mathcal{O}(-a_i) \rightarrow E \rightarrow \mathcal{I}_{C_j}(c + a_j)$, with $C_1 = L$, $C_2 = Q$, vanishes generically, hence everywhere because its target is torsion-free. Thus s_i factors through $\mathcal{O}(-a_j)$. But

$$\mathrm{Hom}(\mathcal{O}(-a_i), \mathcal{O}(-a_j)) = H^0(\mathcal{O}(a_i - a_j)) = 0 :$$

there are no polynomials of negative degree. This would force $s_i = 0$, a contradiction. Thus $d \geq 0$, and the integer equation forces $d = 1$ and $a_2 - a_1 = 1$. Composing either section with the other quotient gives this same determinant in $H^0(\mathcal{I}_L(1))$ and $H^0(\mathcal{I}_Q(1))$, up to sign. Therefore

$$\boxed{L \cup Q \subset H = Z(s_1 \wedge s_2)}.$$

The remaining Chern coefficients give $c = -2a_1$, $c_2(E) = (1 + a_1^2)h^2$ and $c_3(E) = 2h^3$. The last equality again rules out local freeness. These are necessary conditions, not an existence proof for the common sheaf.

How many extensions are available? For $C = L, Q$, Appendix A gives

$$\mathrm{Ext}^1(\mathcal{I}_C(c + a), \mathcal{O}(-a)) \cong H^1(\mathcal{O}_C(c + 2a - 4))^{\vee}.$$

The dimensions are, respectively,

$$\max\{0, 3 - c - 2a_1\}, \quad \max\{0, 7 - 2c - 4a_2\}.$$

Here $L, Q \cong \mathbb{P}^1$, but $\mathcal{O}_Q(1)$ has degree two. Nonzero extension classes need not give locally free middle sheaves.

A Reminder on sheaves and Ext

All sheaves below are coherent. Rank and torsion are taken over an integral variety. A sheaf is *locally free* if locally it is $\mathcal{O}^{\oplus r}$; these are exactly sheaves of sections of vector bundles. Rank is dimension at the generic point, and alone says nothing about local freeness everywhere.

Torsion and reflexivity. Over a domain, m is torsion if $fm = 0$ for some nonzero scalar f . Torsion-free means this forces $m = 0$. For instance $R/(x)$ has torsion over $R = \mathbb{C}[x]$, although it has no nonzero nilpotents. Write $F^* = \mathcal{H}om(F, \mathcal{O}_X)$. The canonical evaluation map is

$$F \longrightarrow F^{**}, \quad s \longmapsto (\varphi \mapsto \varphi(s)).$$

It is injective precisely when F is torsion-free, and an isomorphism precisely when F is *reflexive* [4]. Locally free implies reflexive, which implies torsion-free. For example, $\mathcal{I}_L^{**} = \mathcal{O} \neq \mathcal{I}_L$ for a line in $\mathbb{P}^3$.

Twisting and exactness. We have $F(a)(b) = F(a+b)$ and $\mathcal{H}om(\mathcal{O}(-a), F) = H^0(F(a))$. Tensoring preserves every short exact sequence precisely for a *flat* sheaf. Locally free sheaves are flat: if $F \cong \mathcal{O}^{\oplus r}$, then

$$A \otimes F \cong A^{\oplus r}, \quad a \otimes e_i \longmapsto (0, \dots, a, \dots, 0).$$

Thus tensoring locally makes r copies of the sequence. In particular, twisting by a line bundle is exact. For arbitrary F , tensoring is only right exact: the first injection may be lost.

Global Ext. $\text{Ext}^1(B, A)$ classifies sequences $0 \rightarrow A \rightarrow E \rightarrow B \rightarrow 0$ with end terms fixed; zero is the split extension. Serre duality on $\mathbb{P}^3$ gives

$$\begin{aligned} \text{Ext}^1(\mathcal{I}_C(c+a), \mathcal{O}(-a)) &\cong \text{Ext}^2(\mathcal{O}(-a), \mathcal{I}_C(c+a-4))^\vee \\ &\cong H^2(\mathcal{I}_C(c+2a-4))^\vee \\ &\cong H^1(\mathcal{O}_C(c+2a-4))^\vee. \end{aligned}$$

The second step uses the locally free first argument; the last uses the ideal sequence and intermediate vanishing for $\mathcal{O}(m)$.

Sheaf Ext: local freeness and reflexivity. Unlike global Ext, sheaf Ext records local modules:

$$\mathcal{E}xt^i(F, \mathcal{O})_x \cong \text{Ext}_{\mathcal{O}_{X,x}}^i(F_x, \mathcal{O}_{X,x}).$$

For torsion-free F on a smooth projective threefold, put $\mathcal{E}^i = \mathcal{E}xt^i(F, \mathcal{O})$. Then [3, Section 1.1]:

- F is locally free exactly when $\mathcal{E}^1 = \mathcal{E}^2 = 0$.
- F is reflexive exactly when $\mathcal{E}^2 = 0$ and $\mathcal{E}^1$ has finite support.

Thus for reflexive F , vanishing of $\mathcal{E}^1$ tests local freeness. For a general torsion-free sheaf one must also check second Ext: $\mathcal{E}xt^1(\mathcal{I}_p, \mathcal{O}) = 0$ but $\mathcal{E}xt^2(\mathcal{I}_p, \mathcal{O}) \cong \mathcal{O}_p$.

References

- [1] E. Arrondo, *A home-made Hartshorne–Serre correspondence*, Theorem 1, arXiv:math/0610015.
- [2] R. Hartshorne, *On Rao’s Theorems and the Lazarsfeld–Rao Property*, arXiv:math/0302078.
- [3] D. Huybrechts and M. Lehn, *The Geometry of Moduli Spaces of Sheaves*, Section 1.1.
- [4] The Stacks Project, Tag 0AVT: reflexive sheaves.